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EXTENDED ABSTRACT

Equivariance-based self-supervised learning and SAR tomography for monitoring forest structures

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1. Introduction and objectives

Forests play a fundamental role in the Earth system by regulating major carbon and water fluxes and influencing climate variability and biodiversity. Quantitative information on forest vertical structure, such as canopy height and tree statistics, is essential for estimating above-ground biomass, constraining carbon-cycle models, and supporting ecosystem assessments [Le Toan et al.(2011)]. Yet, obtaining temporally repeatable measurements of three-dimensional forest structure at global scales remains challenging, particularly in regions affected by persistent cloud cover or limited field observations.

Synthetic Aperture Radar (SAR) tomography (TomoSAR) provides a powerful solution to this challenge by enabling the retrieval of the vertical distribution of radar backscatter within forested volumes using multibaseline acquisitions [Aghababaee et al.(2020)]. Its independence from illumination and weather conditions makes TomoSAR especially well suited for large-scale and long-term forest monitoring, including forthcoming spaceborne data from ESA BIOMASS mission, which aim to provide global measurements of forests in support of climate research and carbon-cycle assessments [Le Toan et al.(2011)]. Recent deep learning methods [Berenger et al.(2023)] have demonstrated potential gains in separation and computational efficiency for TomoSAR reconstruction of forests compared to spectral estimators or regularized inversion techniques [Stoica et al.(2005), Aguilera et al.(2013)]. However, the lack of reference data remains an obstacle, especially for supervised learning approaches.

The objective of this work is to address these limitations through a self-supervised deep learning framework for TomoSAR forest structure monitoring, enabling training without external annotations.

2. Methods and data

The proposed approach builds on a SAR tomographic forward model, which relates multibaseline SAR measurements to the vertical distribution of forest reflectivity. For a given resolution cell, the complex measurement vector $\mathbf{y} \in \mathbb{C}^N$ acquired from N baselines is modeled as $\mathbf{y} = \mathbf{A}\mathbf{s} + \boldsymbol{\varepsilon}$, where $\mathbf{A} \in \mathbb{C}^{N \times N_z}$ is the steering matrix determined by acquisition geometry configuration, $\mathbf{s}$ encodes the complex reflectivity vector, and $\boldsymbol{\varepsilon}$ represents the thermal noise [Aghababaee et al.(2020)]. The objective of forest TomoSAR imaging is the estimation of the vertical reflectivity profile $\mathbf{p} \in \mathbb{R}_+^{N_z}$, where $\mathbf{p}_i = |\mathbf{s}_i|^2$.

Rather than relying on unknown ground-truth reflectivity profiles, the method adopts a self-supervised learning strategy inspired by Equivariant Imaging [Chen et al.(2021)]. Coarse input profiles

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are first obtained using beamforming [Stoica et al.(2005)], $\mathbf{p}_i^{(\mathrm{B})} = \frac{\mathbf{a}^{\mathrm{H}}(z_i)\widehat{\mathbf{R}}\mathbf{a}(z_i)}{N^2}$, where $\widehat{\mathbf{R}}$ is the empirical correlation matrix estimated from multi-looked measurements $\mathbf{y}$ and $\mathbf{a}(z_i)$ is the i-th column of $\mathbf{A}$, i.e. the steering vector referring to the noise-free unit amplitude response of a scatterer at height z_i. A lightweight encoder-decoder neural network then maps these input profiles to deconvolved reflectivity profiles, using the learning paradigm presented in Figure 1. The learning objective is defined through a self-supervised loss that combines measurement consistency with equivariance regularization:

$$\mathcal{L} = \underbrace{\left\| \mathbf{Q}_\varepsilon(\widehat{\mathbf{p}})(\mathbf{A}\mathrm{diag}(\widehat{\mathbf{p}})\mathbf{A}^{\mathrm{H}} + \sigma_\varepsilon^2 \mathbf{I}_N)\mathbf{Q}_\epsilon(\widehat{\mathbf{p}}) - \widehat{\mathbf{R}} \right\|_F^2}_{\mathcal{L}_{\mathrm{DAT}}} + \alpha \underbrace{\left\| 10\log(f_\theta(\mathbf{p}_g^{(\mathrm{B})})) - 10\log(\mathbf{T}_g\widehat{\mathbf{p}}) \right\|_2^2}_{\mathcal{L}_{\mathrm{REG}}} \quad (1)$$

with $\widehat{\mathbf{p}} = f_\theta(\mathbf{p}^{(\mathrm{B})})$ the profile predicted by the neural network f_θ, σ_ε^2 the thermal noise level, $\mathbf{Q}_\varepsilon(\widehat{\mathbf{p}})$ the normalization factor to get a correlation $\mathbf{R}$ rather than a covariance matrix, $\mathbf{p}_g^{(\mathrm{B})}$ the beamforming profile computed from measurements simulated with the transformed profile $\mathbf{T}_g\widehat{\mathbf{p}}$, and $\mathbf{T}_g \in \mathbb{R}^{N_z \times N_z}$ the transformation by which the set of signals is invariant [Chen et al.(2021)]. The data term enforces physical consistency with observed correlation matrices, while the regularization term reduces the number of possible solutions by using an equivariant-transformed output profile as a reference for self-supervision.

This approach is evaluated on airborne L-band and P-band TomoSAR datasets from the BioSAR-2 and TropiSAR campaigns, spanning boreal and tropical forest ecosystems [Berenger et al.(2023)].

3. Preliminary results

Figure 1 shows the proposed approach trained for each dataset separately successfully reconstructs vertical forest structure across acquisition geometries and forest types. It improves ground and crown layers separation compared to spectral estimators and is more computationally efficient than iterative methods, supporting scalable large-scale forest monitoring. This framework thus provides a strong foundation for operational TomoSAR processing in forest and climate applications.

4. Discussion

Future work will integrate BIOMASS data, predict forest structure indicators, and generalize across acquisition geometries and forest types without retraining, critical for large-scale forest monitoring.

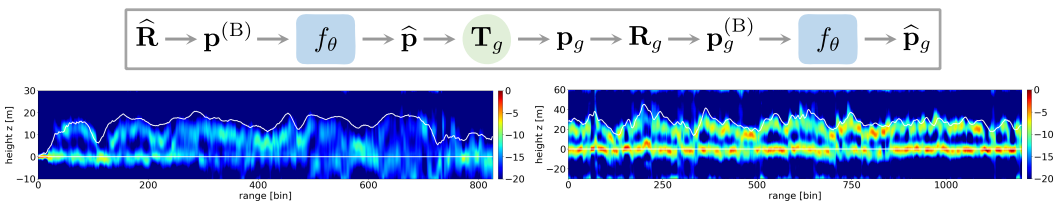

Figure 1. *Schematic overview of the equivariance-based self-supervised TomoSAR learning framework (above), and predicted tomograms (below) on BioSAR-2 (left) and TropiSAR (right) data.*

References

[Le Toan et al.(2011)] Le Toan, T., et al. (2011). The BIOMASS mission: Mapping global forest biomass to better understand the terrestrial carbon cycle. *Remote Sensing of Environment* **115**, 2850-2860.

[Aghababaee et al.(2020)] Aghababaee, H., et al. (2020). Forest SAR tomography: Principles and applications. *IEEE Geoscience and Remote Sensing Magazine* **8**, 30-45.

[Berenger et al.(2023)] Berenger, Z., et al. (2023). A deep-learning approach for SAR tomographic imaging of forested areas. *IEEE Geoscience and Remote Sensing Letters* **20**, 1-5.

[Stoica et al.(2005)] Stoica, P., and Moses, R. L. (2005). Spectral analysis of signals. *Prentice Hall*, Upper Saddle River, NJ.

[Aguilera et al.(2013)] Aguilera, E., et al. (2013). Wavelet-based compressed sensing for SAR tomography of forested areas. *IEEE Transactions on Geoscience and Remote Sensing* **51**, 5283-5295.

[Chen et al.(2021)] Chen, D., et al. (2021). Equivariant imaging: Learning beyond the range space. *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, 4359-4368.